THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An indicator tells us the nature of a solution by: 1

(a) colour change **(b)** smell only **(c)** its weight **(d)** its volume

2. Which indicator changes from yellow to red in a base? 1

(a) China rose **(b)** turmeric **(c)** litmus **(d)** red cabbage only

3. A farmer finds that soil is acidic. He should add: 1

(a) slaked lime **(b)** citric acid **(c)** lemon juice **(d)** vinegar

4. Which statement about neutralisation is correct? 1

(a) it gives only water **(b)** it gives salt and water

(c) it needs two acids **(d)** it needs no acid

5. Which gas is produced when baking soda reacts with an acid like vinegar? 1

(a) hydrogen **(b)** oxygen **(c)** nitrogen **(d)** carbon dioxide

6. A farmer finds that soil is too basic. He may add: 1

(a) lime **(b)** washing soda

(c) baking soda **(d)** organic manure or compost

7. Which of these changes occurs when an acid is neutralised? 1

(a) it becomes more acidic **(b)** it becomes gas

(c) it loses its acidic property **(d)** it becomes solid

8. Which of these pairs is an acid and a base? 1

(a) lemon and curd **(b)** water and salt

(c) soap and baking soda **(d)** vinegar and baking soda

9. Factory waste is acidic. It is treated with: 1

(a) common salt **(b)** sugar **(c)** a base **(d)** an acid

10. Which of the following correctly shows ordering from most acidic to least acidic (generally)? 1

(a) water, lemon juice, tomato juice **(b)** water, tomato, lemon

(c) tomato juice, water, lemon juice **(d)** lemon juice, tomato juice, water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How would you show that lemon juice and tomato juice are acidic? 2

12. Why is dilute acid safer to handle than concentrated acid? 2

13. Why does the red cabbage juice turn red in vinegar? 2

14. How would you show that washing soda is basic? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student adds a base slowly to an acid in a beaker and records the temperature rise. Explain. 3

16. Why do we use lemon juice or vinegar to clean brass? 3

17. Explain the term 'salt' with an example. 3

OR

What would happen if a student adds excess base to an acidic solution after neutralisation? 3

18. Write word equations for nitric acid with potassium hydroxide and hydrochloric acid with calcium hydroxide. 3

OR

Why is toothpaste basic in nature? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss with examples how neutralisation reactions help in solving problems of factory waste, acidic soil, indigestion and insect bites. 5

OR

Explain how an antacid works and why overuse is harmful. 5

20. Explain why acid should be handled carefully, with the safe way of preparing a dilute acid solution. 5

OR

Compare the physical properties of acids and bases (taste, touch, effect on indicators and metals). 5

END OF QUESTION PAPER — Total: 40 Marks

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